**ECE 340** # Homework Assignment #1 **Fall 2026**

DUE: FRIDAY, SEPTEMBER 04, 2026

MAKE SURE YOUR **NAME** AND **NETID** ARE CLEARLY VISIBLE. SUBMIT ASSIGNMENTS AS A PDF (EITHER SCANNED OR TYPED). SHOW ALL WORK (PARTIAL CREDIT CANNOT BE GIVEN WITHOUT WORK). SHOW ALL UNITS. HOMEWORK MUST BE TURNED IN THROUGH GRADESCOPE. HOMEWORK SHOULD BE TURNED IN ON TIME AT 5:00PM ON THE DUE DATE. PLEASE CONTACT THE COURSE DIRECTOR, PROFESSOR DALLESASSE, SHOULD ANY ISSUES WITH LATE HOMEWORK ARISE.

## 1. UNITS AND MEASUREMENTS IN SEMICONDUCTORS

(A). In this course, we will consider often the concept of energy of a photon. A convenient formula for the photon energy in vacuum ($E_{\text{ph}}$), with the knowledge of the wavelength ($\lambda$), is:

$$E_{\text{ph}} = \frac{1.24\,(\text{eV}\cdot\mu\text{m})}{\lambda(\mu\text{m})} = \frac{1240\,(\text{eV}\cdot\text{nm})}{\lambda(\text{nm})} \tag{1.1}$$

Derive this formula, knowing that

$$\lambda = \frac{c}{\nu} \tag{1.2}$$

where $c$ is the speed of light, $\nu$ is the photon frequency, and

$$E_{\text{ph}} = h\nu \tag{1.3}$$

where $h$ is Planck's constant.

(B). Planck's constant can be expressed either in terms of Joules-second (J·s) or of electronvolts-second (eV·s). Reflect on which of the two versions of Planck's constant will be more convenient for your calculations. You can find Planck's constant values in Appendix II of the textbook, in Wikipedia entries, and many other sources. Take the ratio between the values of Planck's constant (in J·s) and the one in (eV·s). What value do you get and what does it represent?

(C). Consider a photon with energy 1.20 eV. What is the wavelength (in nm) of that photon?

(D). Measurements of semiconductor properties are often conducted at reference temperatures that can be easily reproduced at atmospheric pressure (note, you should always use Kelvin units for temperature). Many formulas used in semiconductor theory refer to the so-called "thermal voltage":

$$V_{\text{th}}(\text{eV}) = k_B(\text{eV/K}) \times T(\text{K}) = k_B T \tag{1.4}$$

where $k_B$ is Boltzmann's constant and $T$ is the temperature. Calculate the value of the thermal voltage at the following temperatures:

1. Standard room temperature (300 K).
2. 120 K.
3. Liquid nitrogen temperature (find this temperature value online).
4. Liquid helium temperature (find this temperature value online).

## 2. Lattice and Crystal Structures

Complete the following questions for 1) Simple Cubic (SC) Lattice, 2) Body-Centered Cubic (BCC) Lattice, 3) Face-Centered Cubic (FCC) Lattice and 4) Diamond Crystal Structure.

(A). Calculate the total number of atoms found inside the cubic unit cell. Pay attention to the boundary of unit cell for each case.

(B). Calculate the nearest neighbor distance with the assumption of lattice constant of $a$.

(C). Which structure has the largest atomic density? Which one has the smallest atomic density? Specify the reason.

(D). For Simple Cubic Lattice, Body-Centered Cubic Lattice and Face-Centered Cubic Lattice, calculate the packing fraction (the maximum fractional volume of unit cell that are occupied by sphere (atoms)) of each lattice structure.

## 3. Density of Crystals and Bonding

(A). Calculate the (mass) densities $\left(\frac{\text{g}}{\text{cm}^3}\right)$ of Si and InP from their lattice constants (Appendix III), atomic weights, and Avogadro's number. Qualitatively compare the results with the densities given in Appendix III (no written comparison needed).

(B). For the following commonly used materials determine the bonding type (covalent, mixed ionic-covalent, metallic, and ionic). No explanation needed.

1. Si
2. InP
3. NaCl
4. Au
5. P

## 4. Epitaxial Growth

(A). Define the term Critical Thickness.

(B). Define the term Pseudomorphic Growth.

(C). Define the term Strained Layer Superlattice.

(D). Name one epitaxial growth technology.